

Biology Final — Glossary

Test: Friday, May 29, 2026 · 1–3 PM · North Gym

Every term the teacher listed on the review guide, with a short definition + a one-line "what to remember" cue.

TOPIC I — Foundational Science

Term	Definition
Hypothesis	A testable, falsifiable prediction stated as an "if...then..." relationship between variables.
Independent variable	The one thing you intentionally change. Goes on the X-axis.
Dependent variable	The thing you measure (it depends on the independent variable). Goes on the Y-axis.
Control	The group that does NOT receive the experimental treatment, used for comparison.
Constant	Any variable kept the same across all groups so it can't explain the results.
Bar graph	Use for comparing categorical data (e.g., heights of 4 different species).
Line graph	Use for showing change over time or a continuous trend.
Pie chart	Use for showing parts of a whole (percentages must sum to 100%).

TOPIC II — RNA & Protein Synthesis (Ch. 14)

Term	Definition
Central dogma	DNA → (transcription) → RNA → (translation) → Protein.
RNA polymerase	Enzyme that binds DNA at a promoter and builds the mRNA copy.
Promoter	DNA sequence that tells RNA polymerase where to start transcribing.
mRNA (messenger RNA)	Carries the coded message from DNA out of the nucleus to a ribosome.
tRNA (transfer RNA)	Brings the correct amino acid to the ribosome; has an anticodon that pairs with the mRNA codon.
rRNA (ribosomal RNA)	Makes up the ribosome itself.
Ribosome	The organelle (in the cytoplasm) where translation happens — proteins are built here.
Amino acid	The building block of proteins. 20 different amino acids in humans.
Protein	A chain of amino acids (a polypeptide) folded into a functional shape.
Codon	A 3-base sequence on mRNA that codes for ONE amino acid (or a stop signal).

Term	Definition
Anticodon	The 3-base sequence on tRNA that base-pairs with an mRNA codon.
Start codon	AUG — also codes for methionine (Met). Translation always starts here.
Stop codon	UAA, UAG, UGA — signals the ribosome to release the protein.
Codon wheel	Lookup tool: read mRNA codon from inside → outside to find the amino acid.
Mutation	A change in the DNA sequence.
Substitution	One base is swapped for another. Can be silent / missense / nonsense.
Insertion	A base is added — usually causes a frameshift.
Deletion	A base is removed — usually causes a frameshift.
Frameshift	Insertion or deletion that shifts the reading frame for every codon downstream. Usually very damaging.
Silent mutation	Substitution that still codes for the same amino acid — no effect.
Missense mutation	Substitution that changes ONE amino acid.
Nonsense mutation	Substitution that creates a premature stop codon — protein gets cut short.
Differentiation	The process by which cells become specialized in structure and function (skin cell vs. nerve cell).

TOPIC III — Meiosis (Ch. 12.4)

Term	Definition
Fertilization	When a sperm (n) and egg (n) join to form a zygote (2n).
Zygote	A fertilized egg — the first 2n cell of a new organism.
Homologous chromosomes	Matching pair of chromosomes (one from mom, one from dad) that carry the same genes.
Oogenesis	Formation of female gametes (eggs) — produces 1 egg + 3 polar bodies per meiosis.
Spermatogenesis	Formation of male gametes (sperm) — produces 4 sperm per meiosis.
Gamete	A reproductive cell (sperm or egg). Always haploid (n).
Haploid (n)	A cell with ONE set of chromosomes. Humans: $n = 23$.
Diploid (2n)	A cell with TWO sets of chromosomes (one from each parent). Humans: $2n = 46$.
Tetrad	The 4-chromatid structure formed when homologous chromosomes pair up in Prophase I.
Crossing over	Exchange of DNA segments between homologous chromosomes during Prophase I — a major source of genetic variation.
	Chromosomes that contain new combinations of alleles after crossing over.

Term	Definition
Recombinant chromosomes	
Independent assortment	Homologous pairs line up randomly at Metaphase I, so each gamete gets a random mix of mom's/dad's chromosomes. Another source of variation.

TOPIC IV — Genetics (Ch. 12)

Term	Definition
Genetics	The scientific study of biological inheritance.
Gene	A segment of DNA that codes for a trait, passed from parent to offspring.
Allele	A different version of the same gene (e.g., the "brown eyes" allele vs. the "blue eyes" allele).
Trait	A specific characteristic of an individual (e.g., flower color).
Dominant	The allele that is expressed if at least one copy is present (uppercase letter, e.g., P).
Recessive	The allele only expressed when both copies are present (lowercase, e.g., p).
Homozygous (pure)	Two identical alleles for a gene (PP or pp).
Heterozygous (hybrid)	Two different alleles for a gene (Pp).
Genotype	The genetic makeup (the actual letters, e.g., Pp).
Phenotype	The physical/observable trait (e.g., "purple flowers").
Hybrid	A heterozygous offspring from two genetically different parents.
Punnett square	A grid used to predict the genotypes and phenotypes of offspring from a cross.
Independent assortment	Mendel's law: alleles for different traits segregate into gametes independently.
Incomplete dominance	Neither allele is fully dominant — heterozygote shows a BLENDED phenotype (e.g., red × white = pink).
Codominance	Both alleles are fully expressed at the same time — heterozygote shows BOTH phenotypes (e.g., AB blood, roan cattle).
Multiple alleles	A gene that has more than 2 possible alleles in the population (e.g., human blood: I ^A , I ^B , i).
Polygenic trait	A trait controlled by MULTIPLE genes (e.g., height, skin color) — produces a bell-curve distribution.
Sex chromosome	The X and Y chromosomes (the 23rd pair) that determine biological sex.
Autosome	Any of the 22 non-sex chromosomes.
Sex linkage (X-linked)	A gene located on the X chromosome. Males (XY) only have one X, so they show recessive X-linked traits more often (e.g., color-blindness, hemophilia).

Term	Definition
Karyotype	A picture of all an organism's chromosomes lined up by size and shape. Used to detect chromosomal abnormalities and biological sex.

TOPIC V — Evolution (Ch. 17 & 18)

Term	Definition
Evolution	Change in the heritable characteristics of a population over generations.
Fossil	Preserved remains or traces of an ancient organism. Evidence of evolution.
Artificial selection	Humans breed organisms for desired traits (e.g., dog breeds).
Natural selection	Individuals with traits better suited to their environment survive and reproduce more — passing those traits on.
Adaptation	A heritable trait that increases an organism's fitness in its environment. Can be physical OR behavioral.
Fitness	How well an organism survives AND reproduces in its environment. Fitness = babies.
Survival of the fittest	Individuals with higher fitness leave more offspring. (Note: "fittest" ≠ strongest — it means best-reproducing.)
Descent with modification	Darwin's term — modern species descended from ancestral species with changes over time.
Common ancestry	All life on Earth shares a common ancestor in the distant past.
Homologous structure	Same underlying anatomy, different function (e.g., human arm, whale flipper, bat wing). Evidence of common ancestry.
Analogous structure	Same function, DIFFERENT underlying anatomy (e.g., bee wing vs. bat wing). Evolved independently.
Vestigial structure	An inherited body part with little or no current function (e.g., human appendix, whale pelvic bones).
Allele frequency	How often a particular allele shows up in a gene pool (as a fraction or %).
Gene pool	All the alleles for all the genes in a population.
Genetic drift	Random changes in allele frequency, especially in small populations.
Bottleneck effect	Genetic drift after a population is drastically reduced (e.g., disease, disaster).
Founder effect	Genetic drift when a small group starts a new population — their allele frequencies become the new "normal."
Directional selection	One extreme of a trait is favored — bell curve SHIFTS in one direction.
Stabilizing selection	The middle of a trait is favored — bell curve NARROWS.
Disruptive selection	Both extremes are favored — bell curve SPLITS into two peaks.

Term	Definition
Reproductive isolation	Two populations can no longer interbreed. Required for speciation.
Speciation	The formation of a new species.
Behavioral isolation	Reproductive isolation caused by different courtship rituals (e.g., meadowlark songs).
Geographic isolation	Reproductive isolation caused by a physical barrier (e.g., a canyon separating squirrels).
Temporal isolation	Reproductive isolation caused by reproducing at different TIMES.
Cladogram	A branching diagram showing how species are related by common ancestors.

TOPIC VI — Ecology

Term	Definition
Biosphere	All life on Earth + all parts of Earth where life exists.
Biome	A group of ecosystems with similar climate and organisms (e.g., desert, tundra).
Ecosystem	All the organisms in a place PLUS their physical environment.
Community	All the populations of different species living together in a defined area.
Population	Members of the SAME species living in the same area.
Species	A group of organisms that can interbreed and produce fertile offspring.
Niche	The full role a species plays — where it lives, what it eats, how it interacts.
Habitat	The physical PLACE where a species lives. (Habitat = "address," niche = "job.")
Abiotic	Non-living factors (water, sunlight, soil, temperature).
Biotic	Living factors (other organisms).
Autotroph (primary producer)	An organism that makes its own food from sunlight (photosynthesis) or chemicals (chemosynthesis).
Heterotroph (consumer)	An organism that gets energy by eating other organisms.
Herbivore	Eats plants only.
Carnivore	Eats animals only.
Omnivore	Eats both plants and animals.
Scavenger	Eats already-dead animals (e.g., vulture).
Decomposer	Breaks down dead matter chemically (bacteria, fungi).
Detritivore	Eats detritus — small particles of decaying matter (e.g., worms).
Primary consumer	Eats producers (herbivores).
Secondary consumer	Eats primary consumers.

Term	Definition
Tertiary consumer	Eats secondary consumers.
Food chain	A single linear path of who-eats-whom.
Food web	A network of interconnected food chains.
Trophic level	One step in a food chain/web. Producers are always level 1.
10% rule	Only ~10% of energy at one trophic level passes to the next; the other 90% is lost as heat / used by the organism.
Trophic cascade	A change at one level ripples through the whole food web (e.g., removing wolves changes plant cover).
Symbiosis	"Living together" — a close, long-term relationship between two species.
Mutualism	+/+ — both species benefit (e.g., bee + flower).
Commensalism	+/0 — one benefits, the other is unaffected (e.g., barnacle on whale).
Parasitism	+/- — one benefits, the other is harmed (e.g., tick on a dog).
Competition	Two organisms need the same limited resource. Can be intraspecific (same species) or interspecific (different species).
Predator/prey	Predator hunts and eats the prey. Not symbiotic (it's brief, not a long-term living-together relationship).
Keystone species	A species that has a disproportionately large effect on its community (e.g., sea otters).
Invasive species	A non-native species that causes harm in its new environment.
Succession	The series of predictable changes a community goes through over time.
Primary succession	Succession starting from bare rock — no soil yet (e.g., after a volcanic eruption).
Secondary succession	Succession after a disturbance that did NOT destroy the soil (e.g., after a wildfire). Faster than primary.
Pioneer species	The first species to colonize a barren area (often lichens or mosses).
Climax community	The stable community at the end of succession.
Carrying capacity	The maximum population size an environment can sustainably support.
Limiting factor	Anything that limits population growth.
Density-dependent factor	Limiting factor that gets stronger as population density increases (disease, competition, predation, parasitism).
Density-independent factor	Limiting factor that hits regardless of population size (weather, natural disaster).
Biogeochemical cycle	A cycle that moves matter through the biosphere (water, carbon, nitrogen, phosphorus cycles).
Nutrient limitation	A nutrient whose scarcity limits productivity in an ecosystem.

TOPIC VII — The Human Body

Reproductive System

Term	Definition
Testes	Male gonads (in the scrotum). Produce sperm + testosterone.
Scrotum	External sac that holds the testes outside the body (keeps them cooler).
Penis	Male organ that delivers sperm; also passes urine.
Epididymis	Coiled tube on each testis where sperm mature after being made. Sperm pass through this on the way to the vas deferens.
Seminiferous tubules	Tubes inside the testes where sperm are actually produced.
Seminal vesicle	Gland that adds nutrient-rich fluid to sperm (along with the prostate) to make semen.
Vas deferens	Tube that carries sperm from the testes to the urethra.
Prostate	Gland that produces seminal fluid (nourishes sperm).
Urethra	In males: carries BOTH urine and semen out of the body.
Ovaries	Female gonads. Contain follicles; produce eggs + estrogen + progesterone.
Fallopian tubes	Passageway from ovary to uterus; site of fertilization.
Uterus	Muscular organ where a fertilized egg implants and develops.
Cervix	The lower opening of the uterus into the vagina.
Vagina	The birth canal; also where the penis enters during intercourse.
Placenta	Organ that develops during pregnancy; transfers nutrients + O ₂ from mother to fetus, removes CO ₂ + waste.
Umbilical cord	Connects the embryo/fetus to the placenta.
Amniotic sac	Protective bubble that surrounds the fetus inside the uterus.
Amniotic fluid	Clear yellowish fluid (98% water) inside the amniotic sac that cushions and protects the fetus.
Testosterone	Main male sex hormone — produced in testes. Drives sperm production + secondary sex characteristics.
Estrogen	Female sex hormone — produced in ovaries. Controls oogenesis + uterine lining + secondary sex characteristics.
Progesterone	Female sex hormone from the corpus luteum. Thickens the uterine lining for implantation.
FSH (Follicle Stimulating Hormone)	From pituitary. In females: matures the follicle. In males: stimulates spermatogenesis.
LH (Luteinizing Hormone)	From pituitary. In females: triggers ovulation. In males: stimulates testosterone production.

Term	Definition
Menstrual cycle	~28-day cycle: Follicular phase → Ovulation (~day 14) → Luteal phase → Menstruation (if no fertilization).

Circulatory System (Textbook 27.2)

Term	Definition
Heart	Muscular pump with 4 chambers. Right side = deoxygenated blood; left side = oxygenated.
Right atrium	Receives deoxygenated blood from the body (via vena cavae).
Right ventricle	Pumps deoxygenated blood to the lungs (via pulmonary artery).
Left atrium	Receives oxygenated blood from the lungs (via pulmonary veins).
Left ventricle	Pumps oxygenated blood to the entire body (via aorta). The strongest chamber.
Pulmonary artery	Carries DEOXYGENATED blood from right ventricle → lungs. (The only artery that carries deoxygenated blood.)
Pulmonary vein	Carries OXYGENATED blood from lungs → left atrium. (The only vein that carries oxygenated blood.)
Superior vena cava	Large vein returning deoxygenated blood from the upper body to the right atrium.
Inferior vena cava	Large vein returning deoxygenated blood from the lower body to the right atrium.
Aorta	The largest artery — carries oxygenated blood from the left ventricle to the whole body.
Aortic arch	The curve at the top of the aorta where it bends to head down through the body. (Specifically labeled on the teacher's heart diagram.)
Pulmonary trunk	The single big vessel coming out of the right ventricle that splits into the right and left pulmonary arteries.
Artery	Carries blood AWAY from the heart. Thick, muscular walls. High pressure.
Vein	Carries blood TO the heart. Thinner walls. Has valves to prevent backflow.
Capillary	Tiny vessel (one cell thick) — site of gas/nutrient exchange with tissues.
Pulmonary circuit	Heart → lungs → heart. Picks up O ₂ , drops off CO ₂ .
Systemic circuit	Heart → body → heart. Delivers O ₂ + nutrients, picks up CO ₂ + waste.
Plasma	Liquid part of blood — carries dissolved substances.
Red blood cells	Carry O ₂ via hemoglobin.
White blood cells	Fight infection (immunity).
Platelets	Help blood clot.

Respiratory System (Textbook 27.2)

Term	Definition
Nasal cavity	Where air enters; warms, moistens, and filters air.
Pharynx	The throat — where nasal and oral cavities meet.
Epiglottis	Flap of tissue that covers the trachea when you swallow (keeps food out of the airway).
Larynx	Voice box; contains vocal cords.
Trachea	The windpipe — tube carrying air to the lungs. Lined with cilia.
Bronchi	Two tubes that branch from the trachea, one to each lung.
Bronchioles	Smaller tubes branching from the bronchi.
Alveoli	Tiny grape-like sacs at the end of bronchioles. Site of gas exchange with the capillaries.
Diaphragm	Dome-shaped muscle below the lungs. Contracts to inhale, relaxes to exhale.
Cilia	Hair-like projections on cells lining the airway; sweep out trapped debris and mucus.